Evaluating Descriptive Quality of AI-Generated Audio Using Image-Schemas

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ABSTRACT

Novel AI-generated audio samples are evaluated for descriptive qualities such as the smoothness of a morph using crowdsourced human listening tests. However, the methods to design interfaces for such experiments and to effectively articulate the descriptive audio quality under test receive very little attention in the evaluation metrics literature. In this paper, we explore the use of visual metaphors of image-schema to design interfaces to evaluate AI-generated audio. Furthermore, we highlight the importance of framing and contextualizing a descriptive audio quality under measurement using such constructs. Using both pitched sounds and textures, we conduct two sets of experiments to investigate how the quality of responses vary with audio and task complexities. Our results show that, in both cases, by using image-schemas we can improve the quality and consensus of AI-generated audio evaluations. Our findings reinforce the importance of interface design for listening tests and stationary visual constructs to communicate temporal qualities of AI-generated audio samples, especially to naïve listeners on crowdsourced platforms.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → *Sound and music computing*.

KEYWORDS

AI-generated audio, audio evaluations, listening tests, web-based interactions

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1 INTRODUCTION

Generative algorithms aim to generate novel audio that matches naturally occurring sounds in their descriptive qualities such as realism, naturalness, or plausibility of the sound [14, 16, 18, 19, 39, 54]. Recent studies have emphasized the need for better perceptual evaluation techniques for audio [2, 4]. Automated objective metrics validate audio quality by measuring concepts that can be statistically represented, such as lack of distortion or noise [3, 24, 26, 33, 42, 47, 48, 52]. Such metrics are faster to evaluate, but fail to find meaningful differences between descriptive perceptual measures such as realism or goodness of a morph. Consequently, such perceptual qualities are evaluated using subjective listening tests.

Listening tests aim to measure the perceptual quality of audio samples with respect to a ground truth or each other. In-person listening tests require a considerable amount of the researcher's time and effort and are expensive to set up. Thus, there is an increasing push within the audio deep learning community to move towards crowdsourced platforms such as Amazon Mechanical Turk (AMT) to conduct these tests. Platforms such as AMT are fast-paced, task-based marketplaces where participants optimize the time they spend on a task. Thus, communicating the task instructions and the audio quality under evaluation concisely and clearly becomes increasingly important. AI-generated audio is typically evaluated on quality concepts for sound progression such as the quality of a morph (or how two sounds are interpolated with each other). While such concepts easily translate back to the ability of the algorithm to smoothly disentangle or interpolate within its latent space, non-experts on crowdsourced platforms may find such technical jargons difficult to understand.

Furthermore, for sounds that are not easily recognizable, such as multi-event, noisy audio textures [39, 40, 46], an ideal audio quality description should explicate the complexity observed in the sound space in a human understandable way. Outlining such complex qualities verbosely using language makes for lengthy task instructions which reduce participant interest in such tasks [56] and affects the overall quality of responses. In contrast, for example, image annotation or evaluation tasks often require only a simple 'glance-and-click' action [10]. Our aim in this paper is to assist novice listeners to understand descriptive audio qualities under evaluation by using visual constructs instead of language or words, and in turn help AI-researchers collate better and meaningful responses from such listening tests.

The design of a typical listening test interface involves listening to two or more sounds in comparison to each other or with respect to a reference. As the number of sounds increases (for instance, a MUSHRA test [28] sometimes involves listening and comparing up to 12 sounds with each other) the demands on the listener's audio memory also increases, thus increasing the task's complexity. Recent human computation research on crowdsourcing shows that as task complexity increases, the quality of responses decreases, and participants more frequently abandon such tasks or submit poor quality responses [20, 56]. Thus, another aim of this paper is to design intuitive interfaces using visual constructs to minimize audio task complexity.

Using the metaphors of image-schemas is a promising avenue to visually explicate the descriptive quality of audio and design intuitive interfaces for listening tests. Image-schemas [30] are recurring structures and patterns of our basic sensory-motor experience that are grounded in our embodied interaction (e.g., walking) with our environment. Our inherent sensory-motor capacities of perceiving space and orientation are employed to understand abstract concepts and perform abstract reasoning. For example, when we hear musicians talk about a composition "..it moves from G to A minor to C.." they are applying their sensory-motor understanding of forward-oriented movement to understand chord progressions. Our aim in this research is to apply such image-schemas to the notion of evaluating audio in crowdsourced settings. We can thus formulate our research questions as:

RQ1 How effective are visual constructs such as image-schemas in communicating the descriptive qualities of AI-generated audio?

RQ2 How effective are task interfaces designed using visual constructs such as image-schemas in evaluating AI-generated audio?

We explore these questions by conducting two sets of experiments with 220 participants across different conditions. In the first experiment, we compare the performance of using image-schemas and language-based descriptions to articulate the descriptive audio quality of the goodness or smoothness of a morph. In the second experiment, we compare the performance of interfaces designed using image-schemas and language for the complex task of evaluating the perceptual linearity of control parameters. In both experiments, we investigate the effectiveness of each condition by measuring the quality of responses and consensus amongst participants. Through this paper, we aim to provide future researchers working at the intersection of HCI and audio AI with novel intuitive representations of audio quality, with an application for obtaining crowdsourced evaluation of audio samples. In summary, the main contributions of this paper are:

- An application of visual constructs, image-schema, to perceptually evaluate AI-generated audio.
- An open-sourced configurable front-end framework ('Crowd-Eval-Audio') to set up different listening test workflows on AMT.
- Validation of the effectiveness of using image-schemas to conduct listening tests on a crowdsourcing platform.

We show that directional image-schemas assist in evaluating sound progression in AI-generated general audio better than language alternatives.

2 RELATED WORK

2.1 Visual metaphors for audio

Image-schemas were first introduced by Lakoff and Johnson [34] and later elaborated in [30] as constructs to understand abstract concepts and perform abstract reasoning. Wilkie et al. [55] use the *container* and *source-path-goal* image-schemas (amongst others) while designing music synthesizing interfaces. This was done to improve the user experience and interactions for both experienced musicians as well as novice producers by building intuitive interfaces and reducing the need to possess specialist signal processing domain knowledge. Inspired by this approach, this paper will be grounded in using the *source-path-goal* image-schema to articulate and communicate the descriptive audio quality under test and to design our listening test interfaces. Our approach employs our inherent understanding of the concept of motion to indicate how an AI-generated audio sample progressively interpolates, transitions, or morphs from the start to its end.

2.2 Task design and clarity of instructions

The effect of the quality of task instructions on worker performance and the quality of responses has been rigorously studied by researchers in the context of image annotation and natural language processing (NLP) tasks on crowdsourced platforms. Gadiraju et al. [20] quantify task clarity based on the measures of goal clarity (what is needed to be done in a task) and role clarity (outlining steps to complete it). They survey crowd workers for ratings on multiple tasks and find that while the intrinsic complexity of a task can be high, the cognitive load associated with it can be reduced by a well-structured task presentation. Additionally, workers reported long sentences and complex words as hindrances to task clarity and hence performance. Other researchers extend this idea by identifying clarity flaws in descriptions and building algorithms based on natural language processing to identify such flaws [43].

Similarly, Wu et al. [56] systematically investigate the relationship between descriptive metrics (related to the task instructions and descriptions) and prospective metrics (related to workers' task preferences including worker confidence and enjoyment of the task). They find that though lengthy and descriptive instructions increase worker trust and accuracy in the responses collected, the uptake and subsequently worker interest in such instruction-heavy tasks may be low. A balance between task design, instructions, and the number of examples is needed to achieve better descriptive and prospective metrics. K. Chaithanya Manam et al. [32] formulate a multistage framework for crowdsourced task description refinement, where workers assist in creating high-quality instructions based on prior feedback and cross-worker collaboration. Huang and Wu et al. [25] emphasize the need to decompose complex tasks into sub-components split across the same or multiple participants, thus narrowing each participant's focus singularly to a sub-task to collect more meaningful responses.

While improving the clarity of task instructions should be the central aim of any crowdsourced quality evaluation research, our work looks further into interface design to simplify the task complexity and using different visual representations to better articulate the audio quality under test. Furthermore, most current research on task design and clarity of instructions focuses on annotation tasks or tasks related to natural language texts. This paper highlights the importance of a clear task design for descriptive audio quality evaluation. While there is extensive research that focuses on interface design and representation for audio annotation tasks [7, 10, 41], to the best of our knowledge, there is very little research that discusses the effect of using various intuitive task interfaces and audio representations for the evaluation of descriptive audio qualities.

2.3 Sound perception studies

In the audio domain, the concept of evaluating sounds using crowdsourced platforms is not new. To do this, researchers have extended ITU standards such as MUSHRA [28] to the web. WebMUSHRA [50] brings the lab-based interface to the web utilizing the standard web audio features available currently on all modern browsers. Cartwright et al. [8, 9] show that web-based MUSHRA can be easily used to conduct listening tests on platforms such as AMT to detect intermediate impairments with results matching a lab-based test. Similarly, research on music similarity judgments [35, 44, 49, 53] shows that pairwise comparisons for musical pieces on AMT lead to good results which are comparable to a lab-based setup even when participants in the test had no prior musical knowledge. While these studies provide us a basis for meaningfully conducting listening tests on crowdsourced platforms, they usually analyze pitched sounds or music and do not delve into complex sounds such as audio textures. Furthermore, most studies look at either audio annotation tasks [37, 51], pairwise comparisons, or comparisons against a ground truth. In this paper, we look further into analysing complex tasks to rank order audio samples with respect to references and each other. Additionally, while these studies outline methods to compare audio samples, they do not focus on meaningfully and concisely articulating task instructions or the descriptive quality under test.

Other studies on music tracking tasks showed that visualizing a parameter such as pitch in a live music performance enabled better tracking, especially amongst non-musicians [58]. Our research attempts to build upon this work to produce a better quality of responses from naïve listeners by visualizing the audio quality under test on AMT. Regarding tooling, multiple web-based software tools exist that can be used to conduct listening tests online. Some make use of ITU standards [29, 50] or behavioral experimental design [12] but none focus on visual constructs for audio. The audio-tokens toolkit [15] shares some features with the interfaces we build. While the authors of the toolkit have demonstrated its use in behavioral sciences using speech, its use in evaluating perceptual differences for AI-generated general audio needs further investigation.

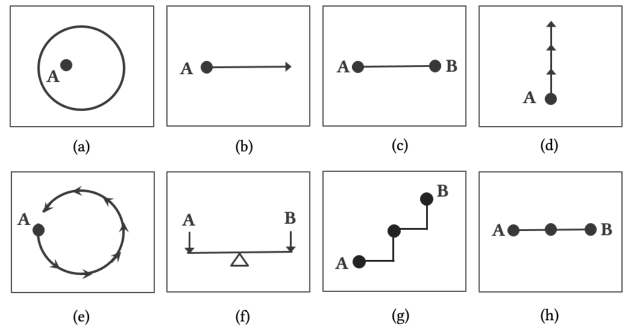


Figure 1: Some image-schemas visualisations which can be used to understand music based on [6, 31, 55] (a) Container, (b) Path, (c) Linkage, (d) Verticality or Up-Down, (e) Cyclic, and (f) Balance. (g) & (h) show two versions of the source-path-goal image-schema used in this paper.

3 METHOD

Our aim with this paper is to use visual constructs of image-schemas to explicate the overall temporal quality of audio in a stationary way (RQ1) and reduce task complexities arising in multi-sample comparisons by designing interfaces based on such constructs (RQ2). We further use both recognizable pitched sounds and unrecognizable multi-event noisy textures in both our experiments to investigate the ability of such constructs to reduce any complexities observed in the generated sound space.

3.1 Image-schemas

Researchers in the audio domain, and particularly in music, have long studied and developed theories of musical meaning based on the metaphors of image-schema [6, 55]. Figure 1 shows some visualizations of common image-schemas used to understand musical concepts. The image-schema of *containment* can be used to understand chords and keys as containers, which can be connected to other chords/keys using different *paths* or *linkages*. Further, we generally perceive notes to go *up* or *down* in pitch. We can also perceive them to sound identical and yet be an octave apart, thus mapping the melodic notes space onto a *cycle* schema. Similarly, the image-schemas for *source-path-goal* can be used to understand a step-by-step melodic progression.

In this paper, we use the *source-path-goal* image-schema to articulate the descriptive audio quality under evaluation and to design listening test interfaces. Generating novel sound morphs is an important task in generative audio modelling [16, 36]. Morphs are generated by interpolating between two points in the latent space or the parameter space of a generative algorithm. Such morph progressions are evaluated for descriptive qualities such as perceptual linearity or interpolation smoothness. Thus, the *source-path-goal* is the most applicable metaphor in our context to indicate a morph progression starting at a selected point in the latent or parameter space (*source*), progressing step-by-step (*path*) towards its end (*goal*).

3.2 Crowd-Eval-Audio framework

We conduct our listening test experiments to explore RQ1 and RQ2 on Amazon Mechanical Turk (AMT). Current affordances on crowd-sourced platforms such as AMT can be limiting for researchers conducting listening tests as:

- Experiments on AMT are usually set up and administered to participants via a simple web page. Such simple pages make it difficult to host a step-by-step workflow required in an online listening test. Workflows are important to guide participants through multiple steps in a listening test, such as the task overview, outlining the consent details, a hearing screening, or post-task surveys.
- The current task administration on AMT does not afford an experimental design where all participants undergo a fixed set of listening trials or a pre-determined number of randomized trials.
- The range of design elements and task widgets available on the platform is limiting. This can lead to un-intuitive and poorly designed interfaces for experiments.

We thus build a front-end-based framework[1] for setting up configurable online listening test workflows on AMT. Our framework needs minimal infrastructure (no database installation required, etc.) and can be easily extended to conduct any type of experiment on AMT. We developed the framework as a single-page application (SPA) using VueJs, Javascript, HTML5, CSS gradients, and Web Audio API. It can be hosted on any content delivery networks (CDNs) and can be administered on AMT using the External Question API[2].

For experiments in this paper, all resources were hosted on Amazon Web Services (AWS) CloudFront distribution network to enable fast downloads of audio files and other resources for crowdsourced participants across the globe. Furthermore, we integrate a participant logging service implemented using AWS Lambda and an Aurora serverless database. This logging was implemented to collate unique responses across all experiments by allowing individuals to attempt tasks from only one experimental condition.

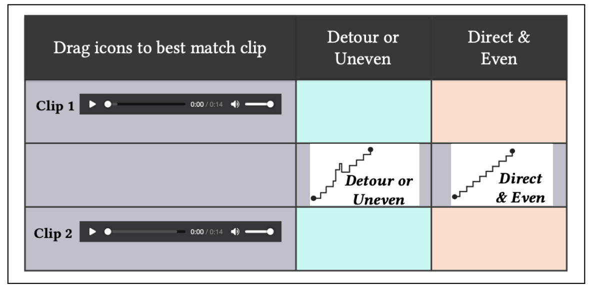
4 EXPERIMENT 1

In this section, we address RQ1 by assessing the effectiveness of using image-schemas to communicate the descriptive audio quality of the 'smoothness or goodness of a morph' of a Generative Adversarial Network's (GAN) [22] latent space in a listening test.

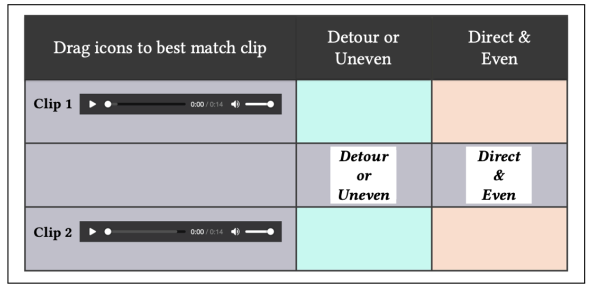
A GAN's latent space is said to be smooth if the sound morphs generated by linearly interpolating between two randomly chosen points in the latent space are perceptually linear. The morphs are said to be non-linear or uneven, if the interpolated sounds jump towards their chosen endpoints quickly instead of progressing gradually, or if the morphs are of a third class of sounds not within the two chosen endpoints in space. Wyse et al. [57] showed that transforming this latent space using Self Organizing Maps (SOM) produces a more even and smoother morph. In this experiment, we conduct listening tests and ask participants to evaluate this smoothness of morph with and without the SOM ordering either using image-schemas or using descriptive language.

[1] Code and sample experiments can be found at - https://github.com/pkamath2/crowd-eval-audio.

[2] https://docs.aws.amazon.com/AWSMechTurk/latest/AWSMturkAPI/



(a) Screenshot of the smoothness of a sound morph experiment using image-schema



(b) Screenshot of the smoothness of a sound morph experiment using language

Figure 2: Experiment 1 listening test user interfaces.

4.1 Study Design

We follow a 2×2 between-subjects factorial design with the following factors -

- Visualization: The use of language or image-schemas to articulate the audio quality under test.
- Type of sounds: The use of pitched sounds or noisy textures.

We recruited $N = 30$ participants for each of the 4 experimental conditions on AMT. Each participant under a condition attempted 3 different trials for pairwise two-alternative forced choice (2AFC) comparisons with randomly selected sounds synthesized from different GANs. A preliminary test (with $N = 3$) was conducted on AMT to analyze and adjust the time allotted and payment for each condition.

4.2 Listening Test Interface

We developed a listening test interface for these experiments consisting of a simple 2AFC type of question. Two image-schemas based on the concept of *source-path-goal* were developed to visualize the descriptive quality of smoothness or goodness of a morph. An image-schema icon for ***"Direct & Even"*** conveys the idea of smooth and even morph. A different icon for ***"Detour or Uneven"*** indicates that the morph is uneven or detours into a third type of sound. Listeners were requested to listen to the two sounds under comparison and drag/drop an image-schema icon near the requisite audio sample. When a participant dragged/dropped one icon toward one audio sample, the interface programmatically moved the other icon toward the other sample under test. This gave the listener visual feedback on how their choice affected the other sound under test in the 2AFC experiment. For the language set of experiments, we outline the descriptive qualities in text without the icons. The language-based interface was designed and implemented to ensure the same functionality as the interface with image-schemas. A schematic of the interface and icons used for both the image-schemas and language conditions are shown in Figure 2. The interfaces developed for this experiment along with their audio samples can be auditioned on our website[3].

4.3 Sound synthesis

To generate morphed sounds for the 2AFC comparisons, we train two GANs - one on pitched sounds from the NSynth dataset [17] and another using a noisy texture dataset [45]. The GANs were trained as outlined in [57]. For pitched sounds, we selected brass and reed instruments from MIDI pitch numbers $64 - 76$ from the NSynth dataset to train the GAN. For textures, we selected 4 different classes of sounds from the texture dataset. To generate the morphs we randomly selected two points A and B in the latent space and sampled 20 evenly-spaced points between them to generate the first clip. Next, the space between the two selected endpoints was remapped using the SOM (as in [57]) and resampled to generate the second clip. Each interpolated point in the latent space generates a stationary morph sample between A and B. Finally, all 20 generated samples are concatenated to create a single audio file which is presented to a listener for evaluation. Each audio sample created in this way is approximately 14 seconds in duration. We generated 3 sets of samples, with and without remapping, for the audio trials in this experiment.

4.4 Participants

The 120 participants recruited for this experiment were paid $1.40 for completing the test. Participants were allowed to attempt our experiments if they had a 95% approval rate with over 1000 approved HITs. The mean task completion times were 7.27 ($SD = 3.26$) and 7.22 ($SD = 3.17$) minutes for image-schema and language-based conditions respectively. Each participant was allowed to complete a task only from one experimental condition.

4.5 Procedure

Participants were requested to sit in a quiet place and use a pair of headphones during the experiment. They were first presented with a hearing screening as outlined in [9]. During the screening, participants were presented with two audio samples containing different tones generated at random frequencies between 55Hz and 10kHz and were asked to count the number of tones. Participants who completed the hearing screening and correctly estimated the number of tones were allowed to attempt the task. The hearing screening ensured that the participants were of normal hearing, were using a pair of headphones, and were in a quiet environment while taking the test.

Next, the participants were presented with the instructions for each condition and were asked for consent. Subsequently, a listening test was presented to the participant depending on their assigned condition. All audio trials within each task were randomized to reduce any ordering effects. After completing the test, they were asked to complete a post-test survey which included questions on their listening equipment and surrounding environment. They were also asked for comments on the complexity of the listening test.

[3]https://animatedsound.com/research/iui2023/crowdaudio/

4.6 Measures for evaluation

The measures outlined below build upon the analysis methodologies from [10, 40].

- **F-Score**: We use F-Scores as a measure of quality for our experiments. F-Score is a harmonic mean of precision and recall and is used as a binary classification measure for this experiment.
- **Pairwise agreement**: We use participant agreement as another measure of quality of responses in our experiments. We build upon the pairwise agreement outlined in [10] to measure the consensus amongst our listeners. This measure can be formalized as:

$$agreement = \frac{1}{N(N-1)} \sum_{i,j=1}^{N} F_{i,j}, \tag{1}$$

  Where N is the total number of participants within a condition and $F_{i,j}$ is the pairwise combined F-score between two participants i, j within that condition.
- **Test-retest reliability**: To estimate the minimum number of participants needed to obtain stable results for our experiments, we measure the test-retest reliability of our results. We randomly divided our participants into two groups and measured Pearson's correlation coefficient between vectors of quality versus condition for each group. Then, we repeated the procedure 1000 times for a range of sample sizes (from 2 to 30) and analyzed the reliability coefficient.

All score distributions, confidence intervals (CI) and standard errors of means are visualized and reported by bootstrap (1000 samples). For every bootstrap iteration, a set of participants equal in number to the participants who took longer than average time to complete each trial were sampled with replacement.

4.7 Results

The responses we collected via our experiments on AMT did not conform to the conditions of normality. We thus conduct non-parametric tests to analyze the results. We first test the stability of our responses using test-retest reliability measures. Our test-retest reliability coefficient for the pitched sounds condition increased from 0.39 at $N = 2$ to 0.95 at $N = 18$. For textures the reliability coefficient increased from 0.24 at $N = 2$ to 0.95 at $N = 24$. Thus, $N = 30$ is sufficient for analyzing all conditions in this experiment.

Effect of visualization on quality of responses. For this experiment, Figure 3 (a) plots F-Score as a measure of quality for the experiment conducted using image-schema ($Mdn = 0.781$) and language ($Mdn = 0.462$) for both pitched and texture sounds. Figures 3 (b) and (c) plots F-Scores individually for pitched sounds ($Mdn = 0.860$ and $Mdn = 0.538$ for image-schemas and language conditions) and textures ($Mdn = 0.727$ and $Mdn = 0.416$ for image-schemas and language conditions) respectively. A Sheirer-Ray-Hare test for quality of response comparison across visualization and sound types, showed that visualizations had significant effect on

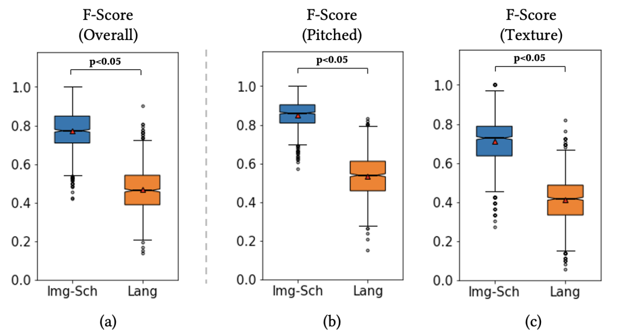


Figure 3: Results of Experiment 1 for pitched and texture sounds based on the type of visualization.

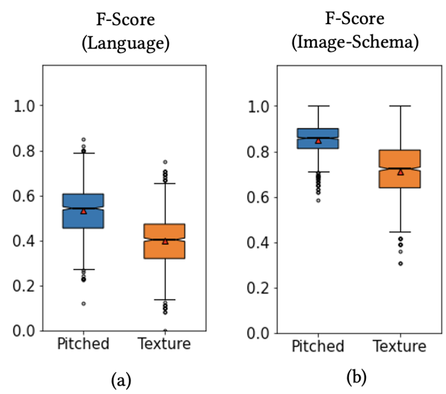


Figure 4: Results of Experiment 1 for language and image-schemas conditions based on sound type.

the overall quality of responses ($H(1, 118) = 20.62, p < 0.017$, adjusted for Bonferroni correction $\alpha = 0.05/3 = 0.017$). A post-hoc Mann-Whitney test found that image-schemas had significant effect on quality of responses for both pitched sounds ($U(N_1 = 30, N_2 = 30) = 674.5.0, p < 0.05$) and textures ($U(N_1 = 30, N_2 = 30) = 637.5.0, p < 0.05$). Therefore, for experiments with simple 2AFC tasks evaluating descriptive qualities of audio samples, using image-schemas significantly improved the quality of responses collected for all sound types.

Effect of visualization for sound types. Figures 4 (a) and (b) plot the F-Scores for pitched sounds and textures based on language and image-schema conditions. Interestingly, even though pitched sounds are generally considered to be more recognizable than noisy textures, participants evaluated both sound types comparably. There were no significant differences found in their evaluation using language ($U(N_1 = 30, N_2 = 30) = 522.5, p = 0.268$) or image-schemas ($U(N_1 = 30, N_2 = 30) = 491.0, p = 0.497$). Therefore, for simple 2AFC type of experiments, complexity in sound space does not significantly affect the quality of responses submitted in an online listening test within the same visualization condition.

Table 1: Pairwise participant agreement for Experiment 1.

Sound Type	Experimental Condition	Median Agreement [95% CI]
Pitched Sounds	Image-Schema	0.897 [0.850, 0.944]
	Language	0.549 [0.502, 0.624]
Textures	Image-Schema	0.803 [0.707, 0.898]
	Language	0.384 [0.292, 0.515]

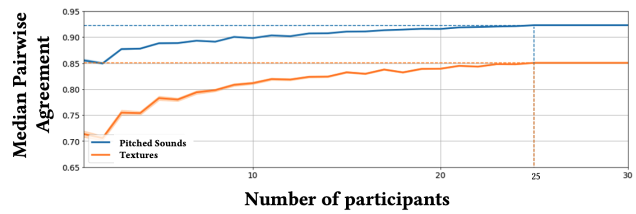


Figure 5: Median pairwise agreement for pitched sounds and textures under the image-schemas condition as the number of participants increases for Experiment 1. The dotted lines indicate the number of participants needed to reach full agreement for this condition.

Pairwise Agreement. Table 1 shows the median pairwise agreement between participants across trials in this experiment. The results indicate that experiments using image-schemas showed significantly better agreement as compared to language-based conditions.

To examine the effect of the number of participants on the overall agreement in this experiment, we plot the aggregated median pairwise agreement by simulating an increase in the number of participants for the image-schemas condition as shown in Figure 5. We adapt this method from [10]. First, we shuffle the order of the responses from all participants across all trials and then progressively add the responses to calculate the overall median agreement for each sound type and repeat the process 1000 times. The dotted lines show the number of participants needed for maximum consensus for each condition. The plots indicate that we need at least 25 participants to reach the maximum consensus for that condition in this experiment for both pitched sounds as well as textures. This implies that for simple 2AFC tasks, especially in the absence of ground-truth labels, the number of participants needed to evaluate sounds with a significant agreement, does not depend upon the complexity of the sounds under test.

Relationship between task time and quality of responses. We study the effect of time taken by participants to complete a task on the quality of responses submitted. For this, we first sort the participant responses in an increasing order of time taken to complete the trials. We then analyse the correlation between the time spent on the trials with the average quality (F-Score) of the responses. For pitched sounds, under both visualization conditions, time taken to complete the task strongly correlated with the quality of responses submitted ($r(28) = .717, p < 0.05$ for image-schemas condition and $r(28) = .819, p < 0.05$ for the language condition).

For textures under the language condition, time spent on task negatively correlated with quality ($r(28) = -.817, p < 0.05$). For the image-schema condition, these sounds moderately correlated with quality ($r(28) = .407, p < 0.05$). Therefore, for recognizable pitched sounds, participants who spent a longer time on a task were able to submit better responses. For unrecognizable noisy textures, participants who spent a longer time on the task submitted lower quality responses when using the language condition as compared to the image-schemas condition. This implies that staying longer on the task evaluating textures might have increased the cognitive load and confusion amongst the participants leading to poorer responses when using the language based conditions, which was slightly alleviated when using image-schemas.

5 EXPERIMENT 2

In this section, we address RQ2 by using image-schemas to design interfaces for a listening test. We conduct an experiment to evaluate the perceptual linearity of sounds generated by linearly varying control parameters of a generative model. Controllable or conditional synthesis is an important task in generative audio modeling which typically comprises of a deep neural network trained on audio data in conjunction with some conditional parameters. For example, generative models conditioned on pitches from different musical instruments can be explicitly controlled to generate different timbres for a pitch [16]. Generally, control parameters used to train such models are varied linearly in the parameter space. Their effect, however, in the generated audio is not always guaranteed to be perceptually linear [1, 23, 39]. There is thus a need to evaluate the perceptual sensitivity to the linear parametric variation of such AI-generated sounds.

5.1 Study Design

As with Experiment 1, this experiment applies the same recruitment criterion and follows a 2×2 between-subjects factorial design using the factors of visualization and type of sounds. Once again, a preliminary test (with $N = 3$) was conducted to analyze and adjust the time allotted and reimbursements for this experiment.

5.2 Listening Test Interface

To conduct this experiment, we designed and implemented an interface to rank-order and proximally space some audio samples. The goal of this test was to present 7 short audio samples - 2 references and 5 sounds under test - and ask the listeners to order the 5 samples perceptually between the two references. For the image-schema condition, a listening test interface followed the *source-path-goal* schema as shown in Figure 6 (a). All audio samples were visually represented as thumbs along a *path* in between the *source* and *goal* reference sounds. A screenshot of the user interface using image-schemas is shown in Figure 6 (b). Black thumbs represented the reference endpoints at the left and right extremes. Colored thumbs along a slider represented the remaining sounds under test. The participants were asked to hover their mouse pointers over the thumbs to listen to the sound samples and to drag the thumbs along the slider to position them with respect to the static and immovable references as well as each other. The 'hover to listen' and 'drag to position' interactivity were implemented to reduce fatigue amongst

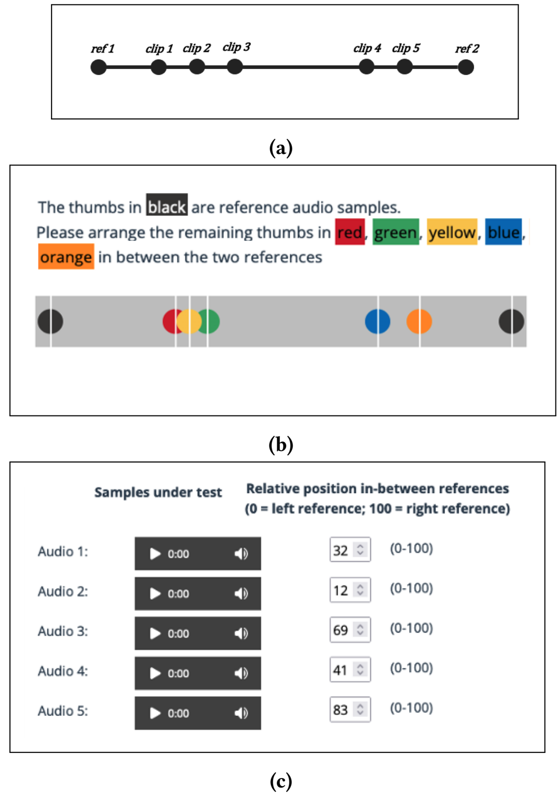


Figure 6: Experiment 2 image-schema design is shown in (a), and the screenshot of the implemented interface using image-schemas is shown in (b). A language-based interface was implemented as shown in (c).

listeners from having to click a web audio component to play each sound multiple times while positioning the sounds between the references.

Given the number of comparisons participants needed to perform, we split this task into 3 sub-tasks to mitigate its complexity. First, the participants were asked to audition the sounds and position them as discussed above. Then, they were given an opportunity to listen to the 'arrangement' they created by clicking a button that played each audio sample in the order in which they were positioned along the slider. This step was intended to aid the participants in positioning the samples better by listening to them in an automated sequence. After this step, they were asked to listen to the arrangement again and fine-tune each sample's position based on not just the reference endpoints, but also based on their proximity to their adjacent neighbors. Participants were allowed to submit the task only after completing all steps. The interfaces developed for this experiment along with their audio samples can be auditioned on our website[4].

The language-based interface, shown in Figure 6 (c), was designed and implemented to ensure the same functionality as the screens with image-schemas. Here, participants listened to audio

[4]https://animatedsound.com/research/iui2023/crowdaudio/

samples presented using the HTML5 web audio interface and entered the ranks or distances using number drop downs between 0 and 100. The two rank-order and proximity based on distance 'arrangement' sub-tasks (just as in the image-schema trials) played the audio samples in an automated sequence based on the order indicated by the numbers in the drop downs.

5.3 Sound synthesis

To explore RQ2 with varying sound complexities, we select two sets of audio samples. The pitched samples were generated using the Syntex [59] DS_BasicFM_1.0 synthesizer which generates a frequency modulated sine wave governed by the algorithm $y[t] = sin(2\pi * cf * t \;+\; mI * sin(2\pi * mf * t)))$, where cf is centre frequency, mI is the modulation index and mf is the modulation frequency. To create this dataset, we fix the modulation frequency and modulation index to $\sim 7.5Hz$ and 12.5 respectively, and linearly vary the signal's center frequency between $\sim 330Hz$ and $\sim 660Hz$ (the left and right references respectively). The center frequencies for the remaining 5 other samples were randomly selected between the two references. Each of these 7 audio samples was 2 seconds long.

For texture samples, we used a recorded sound made by water filling a container where the control parameter guiding the progression of the audio samples was the container's fill-level. The left and right references were chosen with fill-level=0 (empty container) and fill-level=1 (full container). The 5 other samples under test were random fill-levels between the two references.

5.4 Participants & Procedure

We recruited $N = 25$ participants for the 4 experimental conditions resulting in 100 participants. They were paid \$1.20 for completing the task. Each task had one perceptual ordering trial. The mean task completion times were 6.63(SD = 2.23) and 7.31(SD = 3.5) minutes for image-schema and language-based conditions. As in Experiment 1, participants were allowed to complete a task only from one experimental condition.

5.5 Measures for evaluation

We analyze the results from this experiment by first transforming the proximity or distance-based values captured from the participant's responses to ranked data. We use the measure of F-Score and agreement (formalized in section 4.6) on this ranked data to evaluate the quality of responses. In addition to these measures we use -

- **Cosine Similarity Score**: The rank-based F-Score fails to account for the variance in the actual distance values captured (for e.g., while a participant's response of [1,2,3,4,5] matches in rank to a ground truth of [11,28,32,49,51], it differs largely in the actual proximal distance/spacing captured). To capture this variability in distance, we treat each participant's response as a 5-dimensional vector and find its cosine similarity with the ground truth. Cosine similarity for two vectors is formalized as -

$$similarity\ score = \frac{x \cdot y}{\|x\|\|y\|} \tag{2}$$

where x and y are the participant's response and ground truth. The numerator is their dot product and the denominator is a scalar product of their Euclidean norms.

Table 2: Pairwise participant agreement for Experiment 2.

Sound Type	Experimental Condition	Median Agreement [95% CI]
Pitched Sounds	Image-Schema	0.896 [0.850, 0.941]
	Language	0.813 [0.731, 0.859]
Textures	Image-Schema	0.630 [0.476, 0.783]
	Language	0.498 [0.417, 0.651]

5.6 Results

As with Experiment 1, we use non-parametric tests to analyze our data for this experiment. Our test-retest reliability coefficient for the pitched sounds condition in this experiment increased from 0.18 at $N = 2$ to 0.95 at $N = 22$. For textures the reliability coefficient increased from 0.25 at $N = 2$ to 0.95 at $N = 18$. Thus, $N = 25$ is sufficient for analyzing all conditions in this experiment.

Effect of visualization on quality of responses. For this experiment, Figure 7 (a) and (b) plots aggregated rank-based F-Scores for the responses for each sound type. Overall, the experiments with image-schemas did not result in significant differences using rank-based measures ($Mdn = 0.82$ for pitched sounds and $Mdn = 0.58$ for textures) in comparison with language-based experiments ($Mdn = 0.72$ for pitched sounds and $Mdn = 0.5$ for textures). This implies that most participants in both image-schemas and language-based conditions could rank-order the audio samples correctly.

To further study the difference between the two conditions, we tested our hypothesis using distance-based similarity measures defined in section 5.5. Figure 7 (c) and (d) plots this similarity measure for each sound type. A Sheirer-Ray-Hare test for quality of response comparison across visualization and sound types, showed that visualizations had significant effect on the overall quality of responses ($H(1, 98) = 7.13, p < 0.017$), adjusted for Bonferroni correction $\alpha = 0.05/3 = 0.017$). In post-hoc Mann-Whitney tests, both pitched sounds ($U(N_1 = 25, N_2 = 25) = 400.5, p < 0.05$) and textures ($U(N_1 = 25, N_2 = 25) = 410.0, p < 0.05$) report significant differences in evaluation when using image-schemas. Thus, interfaces designed using image-schemas were better at capturing differences in spacing or proximity of the sound samples than language-based interfaces.

Effect of visualization for sound types. Figure 8 shows the rank based F-Score and distance based similarity measures for pitched and texture sounds for both visualization conditions. Even though the texture sounds used in this experiment (water filling a container) were recognizable and not very noisy, pitched sounds were evaluated significantly better than textures when using both rank-based data ($U(N_1 = 25, N_2 = 25) = 404.5, p < 0.05$) as well as distance-based similarity scores ($U(N_1 = 25, N_2 = 25) = 430.0, p < 0.05$).

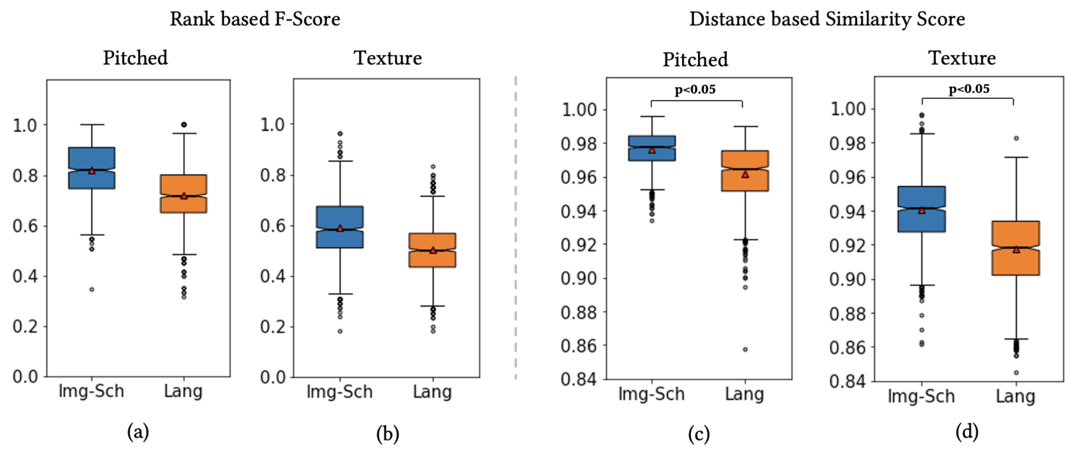


Figure 7: Results of Experiment 2 for pitched and texture sounds based on the type of visualization.

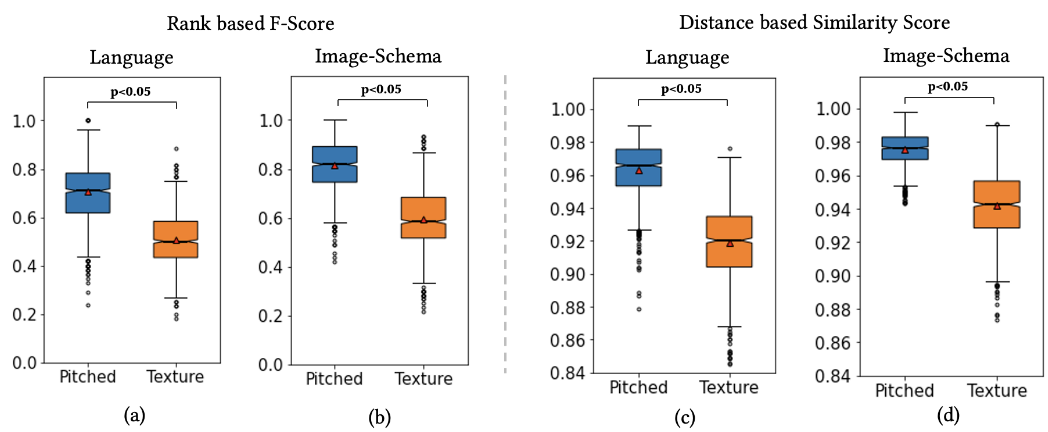


Figure 8: Results of Experiment 2 for language and image-schemas conditions based on sound type.

Pairwise Agreement. Table 2 shows the median agreement for each condition for this experiment. The conditions using image-schemas for both pitched sounds and textures reported significantly better agreement amongst participants than using language. $U(N_1 = 25, N_2 = 25) = 514.0,\ p < 0.05$ and $U(N_1 = 25, N_2 = 25) = 421.0,\ p < 0.05$ for pitched sounds and textures respectively. As in Experiment 1, we also analyze the effect of the number of participants on the overall aggregated agreement for the image-schema condition. For pitched sounds, the maximum consensus in this experiment is reached by aggregating agreement from 9 participants. For textures, we needed 18 participants for maximum consensus. Therefore, for complex tasks, the number of participants needed for consensus varies depending upon the type of sounds under test. Recognizable pitched sounds need fewer participants as compared to textures.

6 DISCUSSION

6.1 Use of image-schemas for audio quality description

We explored RQ1 by conducting an experiment with a simple 2AFC type of task and using pitched sounds and textures of longer duration ($\sim$ 14 seconds). We find that by visually articulating the descriptive audio quality under test, we improve the overall quality of responses collected via listening tests for these conditions. By using image-schemas, we employ the use of conceptual models [30] which are both language and experience agnostic and thus provide means to understand audio quality concisely. These findings reinforce previous research on the clarity and brevity of instructions [20, 56] and extends it to audio evaluation.

We term the pitched sounds used in this experiment as more recognizable than noisy textures as they had some semantic meaning associated with them. For instance, the timbre and/or MIDI pitch of the pitched sounds used in this experiment changed as the sounds progressed in time. In section 4.7 we saw that such pitched sounds were evaluated comparably to noisy textures under both visualization conditions. This implies that the quality of responses in a simple 2AFC type of experiment does not depend on the sound space complexity introduced due to the type of sound being evaluated in a test.

6.2 Use of image-schemas for designing listening test interfaces

We explored RQ2 by conducting an experiment with a complex rank-ordering and proximity spacing type of task and using pitched sounds and textures of shorter duration ($\sim$ 2 seconds). It is not surprising that the interfaces designed using image-schemas were able to collect better responses as they were better designed than the language based conditions. Interestingly though, the participants using both interfaces could rank-order the sounds equally well. The advantage image-schemas condition had over language based interfaces was its ability to allow participants to capture proximal distances between the sounds better. This can be attributed to the fact that the slider-based image-schema interface allowed for better visual positioning of the sounds under test in relation to each other as compared to the number drop downs in the language condition. Therefore, it is sufficient to use language based interfaces for simple rank-ordering tasks and to use image-schema based interfaces for more granular proximity spacing tasks.

In this experiment, participants evaluated pitched sounds better than textures under both conditions. This could be attributed to the fact that textures have more perceptual feature dimensions that could potentially be used for evaluation compared to pitched sounds. For instance, for the texture sample of water filling a container, several other perceptual features, such as the rate of the water filling, the material and the resonances of the bucket etc., co-vary with the fill-level [21] confounding the parameter under evaluation. Furthermore, most participants on AMT are novice listeners and may be more intuitively familiar with pitched sounds than multi-event textures. Also, they may be more comfortable working on tasks with speech transcription, sound event detection, or music evaluation and less familiar with rank ordering tasks for textures, thus lowering the quality of responses collected.

While in Experiment 1, the number of participants needed for maximum agreement did not depend upon the sound type, in Experiment 2, pitched sounds needed fewer participants for maximum agreement than textures (see Section 5.6). This finding has practical implications when selecting the number of participants in a listening test. Thus, when task complexity is high in a listening test, larger sample size is needed for unrecognizable multi-event soundscapes to match in agreement with recognizable pitched sounds evaluated under similar conditions.

Measuring perceptual linearity on a continuous scale, as done in Experiment 2, can be modeled as discrete pairwise comparisons as suggested by prior work in other domains [11, 13]. It should be noted that while discrete pairwise comparisons are easier to evaluate, we are unable to record some important perceptual distance based differences between the samples under evaluation. Thus, for AI-generated audio, we find pairwise comparisons are great for rank-ordering tasks and interfaces designed using image-schemas are advantageous for more granular perceptual distance based evaluations.

6.3 Amazon's Mechanical Turk as representative crowdsourced platform

Besides AMT, the audio AI community uses crowdsourcing platforms such as Zooinverse[5] and Crowdflower[6] to conduct audio evaluations. Each platform's affordances drive the type of experiments that can be conducted and the quality of data collected. For instance, Zooinverse is a research-focused volunteer-driven platform, where participants' motivation to contribute may not be driven by monetary incentives [7]. Comparatively, many participants on platforms such as AMT or Crowdflower are driven by the payment structure associated with the task as income generation is their primary motivation [5, 27, 38]. Furthermore, Zooinverse-like platforms connect researchers and participants more intrinsically via their project discussion boards which assists in a deeper connection and communication on the research compared to the asynchronous email-based communication afforded by AMT. Projects set up on Zooinverse are usually long-running and undergo a review by the Zooinverse team and are assigned an expert to assist researchers with best practices for setting up their experiments. Such affordances potentially allow for better overall quality of responses from volunteer-driven platforms than paid crowdsourcing.

We choose AMT to conduct our experiments because it supports short-term evaluation tasks and the convenience of experiment administration and setup. Though AMT does not afford between-subjects experimentation out-of-the-box, we set up our listening tests to log participant actions and disallow them from attempting more than one task. In addition, we explore communicating tasks and instructions, acknowledging asynchronous communication between participant and researcher. Our findings should hold even in a collaborative environment provided by platforms such as Zooinverse.

7 LIMITATIONS AND FUTURE WORK

7.1 Use of modern browser features

All experiments in this paper rely on the availability and use of modern internet browser features such as HTML5 components, drag/drop, sliders, CSS gradients, and variables. Without these features, experiments with the image-schema conditions will not render correctly. Therefore, before participants started working on our tasks in our experiments on AMT, we performed browser checks and allowed only participants who used compatible browsers to attempt our trials. Given the diverse pool of crowdsourced participants with varying desktop and browser installations, it is difficult to ascertain how many qualified participants could not attempt our tasks because of this limitation.

7.2 Generalizability to other datasets

There are some limitations to generalizability due to the datasets we use in our experiments. In this paper, we choose a set of sounds from varied sources - pitched instrument sounds from an existing benchmarked dataset [17], noisy textures [45], a set of programmatically synthesized pitched sounds [59] and recorded sounds (water filling a container). While we try to cover multiple sources

[5] https://www.zooniverse.org/
[6] https://www.crowdflower.com/

of sounds - from noisy textures to sounds with events and pitched instrument sounds - more experimentation with larger datasets is needed to generalize these results to other sounds systematically.

7.3 Development of image-schemas

In this research, we explore using the *source-path-goal* image-schema for articulation and interface design as both experiments evaluate sound progression. There is a need to design and develop new icons and interfaces depending on the evaluation measure in a listening test. Further, more exploratory research is necessary for evaluating AI-generated audio qualities of stationary sounds using other visual metaphors such as *containment* or *linkage* [30, 34, 55] etc. A potential future avenue for research surrounds the development of a generalized, pluggable web interface or visual library for metaphors for such audio evaluations.

8 CONCLUSION

In this work, we highlight the importance of using the visual metaphors of image-schemas in designing listening test interfaces for AI-generated audio. We introduced novel visual constructs to evaluate sound progression in rank ordering and pairwise comparison types of tasks and verified their effectiveness in improving the overall quality of responses in a listening test. Furthermore, we discuss the implications of using such visual constructs to evaluate sounds with varying complexities. Our findings shed light on the value of using such stationary constructs to communicate the temporal quality of audio for future researchers working at the intersection of generative audio modeling and human-computer interaction.

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